# NEW CAPSTONE PROJECT FORMAT
# INFORMATION TECHNOLOGY/INFORMATION SYSTEM

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CHAPTER I - **INTRODUCTION**

1.1 Project Context

1st Paragraph – general statement on technology related to the study
2nd Paragraph - Connect the technology to the current process of your field
3rd Paragraph - Gaps or the problems of the Organization or society based on survey conducted

Gaps → SOs → technical background → testing

4th Paragraph – Motivation/ Drive why you conducted the study

1.2 Purpose and Description

1ST Paragraph – purpose of the study
2nd Paragraph – Description of the Application (Describe the user accounts, its access control/policy, etc) include who serve as admin of the study.

1.3 Objectives

This section states what the project intends to accomplish. It presents the overall direction of the study or system development and clarifies the expected results. The objectives must be aligned with the identified problem and should be measurable, achievable, and relevant to the project's purpose.

1.3.1 General Objective

It expresses the broad primary goal of the project. It summarizes the main purpose of developing the system, application, or solution. It answers the question:

What is the main goal of this project?

Example structure:

To design and develop a [system/application] that addresses [identified problem] of [organization/community].

1.3.2 Specific Objective

Break down the general objective into smaller, measurable actions or deliverables. These outline the concrete steps needed to achieve the overall goal, such as designing modules, developing features, conducting testing, and evaluating system performance. They answer the question:

What specific tasks must be completed to achieve the general objective?

Specific objectives usually begin with action verbs such as:

- To analyze
- To design
- To develop
- To implement
- To test
- To evaluate
- To deploy

Sample Specific Objectives

- To analyze the existing process of data collection and record management in the organization.

- To design a computerized system architecture including database structure and user interface.

- To develop a web-based system that allows efficient encoding, updating, and retrieval of records.

1.4 Significance of the Project

This section explains the importance of the project, highlighting how it benefits the organization, users, and student-developers while contributing to the achievement of relevant Sustainable Development Goals (SDGs), such as promoting quality education, innovation, digital transformation, and sustainable community development.

1.5 Scope and Delimitation

This section defines what the project will cover, including its main features, coverage, and implementation boundaries, as well as the specific limitations of the system.

## CHAPTER II REVIEW OF RELATED LITERATURE/ SYSTEM TECHNICAL BACKGROUND

This chapter reviews existing systems and studies related to the project to identify best practices and gaps. It also presents the technical background, including the software, hardware, and tools used in system development. This serves as the foundation for designing and implementing the proposed system effectively.

## CHAPTER III METHODS

### 3.1 Model/SDLC

This section describes the Software Development Life Cycle (SDLC) model used in developing the system (e.g., Waterfall, Agile, or Spiral). It explains the phases followed—from planning, analysis, design, development, testing, deployment, to maintenance—to ensure a structured and systematic system development process.

### 3.2 Data Gathering

This section explains how the proponents collected relevant information from the target users or organization through interviews, surveys, observations, and document review. The gathered data served as the foundation in identifying system problems and improvement areas.

### 3.3 Requirements Analysis

This section presents how the proponents analyzed the collected data to determine the system's functional and non-functional requirements. It defines what the proposed system should accomplish and the performance standards it must meet.

### 3.4 Requirements Documentation

This section contains the formal documentation of the approved system requirements, including detailed system features, user roles, and technical specifications. This serves as the official reference throughout the development and implementation of the capstone project.

### 3.5 Design of Software, Systems, Product, and or processes

#### 3.5.1 Conceptual framework

The conceptual framework must be anchored to IPO, include the role of technology in the process.

Figure 1 depicts the overview of the application which anchored in IPO

#### 3.5.2 Detailed Design

3.5.2 Detailed Design - describes the internal workings of the system, covering all user accounts, workflows, and processes from login to exit. It includes diagrams for clarity and technical discussion for each functional module.

##### 3.5.2.1 System Overview Diagram

A high-level system architecture diagram shows the major components of the system, how users interact with it, and how data flows

##### 3.5.2.2 Account-Based Flow Diagrams

3.5.2.2 Account-Based Flow Diagrams - it includes the Admin Account Diagram and technical discussion for each

3.5.2.3 Standard/User Account – it includes the user Account Diagram and technical discussion for each

3.5.2.4 System Process Flow (Login to Exit) – it includes the Step-by-Step Technical Flow from log in to exit

Note: Process flow presentation using flowchart and other appropriate presentations diagrams

3.6 Algorithms

3.7 Capstone Element

3.7.1 Software Requirements & Development

3.7.2 Implementation

3.7.3 Testing and Validation

**CHAPTER IV RESULTS AND DISCUSSION**

This chapter presents the results of the system development, testing, and evaluation of the proposed capstone project. It discusses the implementation of the system based on the identified requirements and demonstrates how the objectives of the study were achieved.

The results include system outputs, screenshots of major modules, performance testing outcomes, and user acceptance testing results. It also presents the evaluation of the system in terms of functionality, usability, reliability, efficiency, and security.

The discussion interprets the findings by explaining how the developed system addressed the identified problems of the organization. It highlights the effectiveness of the system, improvements observed after implementation, and feedback gathered from users during testing.

Overall, this chapter validates that the proposed system meets the specified requirements and contributes to improved operational efficiency and service delivery within the target organization.

**CONCLUSION AND RECOMMENDATION**

**APPENDICES MAY INCLUDE THE FOLLOWING**

- Relevant Source Code
  - ✓ Evaluation Toot or Test Documents
  - ✓ Sample input/output/Reports
- Users Guide
  - ✓ Process/Data/information Flow
  - ✓ Screen layouts
- Test Results
  - ✓ Sample Generated Outputs
  - ✓ Pictures showcasing the data gathering, investigation done (e.g. floor plan, layout, building, etc.)
- One-Page Curriculum Vitae per team member

**SUGGESTED TOPICS FOR CAPSTONE**

Information Technology

4.2.1 Software Development

- Software Customization
- Information Systems Development for an actual client (with pilot testing)
- Web Applications Development (with at least alpha testing on live servers)
- Mobile Computing Systems

4.2.2 Multimedia Systems

- Game Development
- e-Learning Systems
- Interactive Systems
- Information Kiosks

Network Design and Implementation and Server Farm Configuration and Management

IT Management

- IT Strategic Plan for sufficiently complex enterprises
- IT Security Analysis, Planning and Implementation